Combinations to 10 with the Number Rack

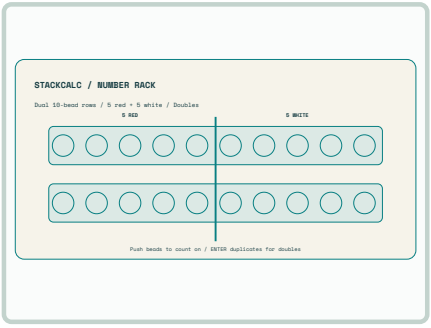
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.K.OA.A.4

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Number Rack

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Push 7 beads to the left in one push. Count white beads remaining on the right.

Sketch Model on Paper:
Color 5 red and 2 white beads pushed to the left on a number rack drawing.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE:	123	Base-10 Arithmetic	
1.	10	X: 10	Key wire total (10)
2.	ENTER	Y: 10 | X: 10	Stack lift copies 10
3.	6	Y: 10 | X: 6	Key pushed beads int
4.	-	X: 4	Subtract: 10 - 6 = 4

EXPECTED LCD X: 4 beads left

1 **Push 8 beads left. How many remain on right?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **Push 4 beads left. How many remain on right?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **Push 5 beads left. How many remain on right?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **Push 9 beads left. How many remain on right?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Unit 3

Combinations to 10 with the Number Rack • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Number Rack (Rekenrek 20-Bead Stand)

CAD Blueprint: number_rack

Target Device: SC-6 Elementary

CAD / STL Source File:

Education/scad/number_rack_parts.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

White Frame with Signal Red & White Beads (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Push 8 beads left. How many remain on right?** Standard: CCSS.MATH.K.OA.A.4	2	[1][0] [ENTER] [8] [-]
2	**Push 4 beads left. How many remain on right?** Standard: CCSS.MATH.K.OA.A.4	6	[1][0] [ENTER] [4] [-]
3	**Push 5 beads left. How many remain on right?** Standard: CCSS.MATH.K.OA.A.4	5	[1][0] [ENTER] [5] [-]
4	**Push 9 beads left. How many remain on right?** Standard: CCSS.MATH.K.OA.A.4	1	[1][0] [ENTER] [9] [-]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Encourage students to slide beads in groups rather than one by one.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.